

Answers learning task 20

2026 – 2027

Exercise 11.70: The data in the table below gave x = the water content of snow on April 1 and y = the yield from April to July (in inches) on the Snake River watershed in Wyoming for 1919 to 1935. (The data were taken from an article in *Research Notes*, Vol. 61, 1950, Pacific Northwest Forest Range Experiment Station, Oregon.)

x	y	x	y
23.1	10.5	37.9	22.8
32.8	16.7	30.5	14.1
31.8	18.2	25.1	12.9
32.0	17.0	12.4	8.8
30.4	16.3	35.1	17.4
24.0	10.5	31.5	14.9
39.5	23.1	21.1	10.5
24.2	12.4	27.6	16.1
52.5	24.9		

- (a) Estimate the correlation between Y and X .
(b) Test the hypothesis that $\rho = 0$ using $\alpha = 0.1$.

Answer

(a) 0.9332

(b) $t_0 = 10.0578 > C_t = 1.7531, p = 0.0000 \leq \alpha = 0.1 \implies \text{reject } H_0$

Exercise 11.72: The weight and systolic blood pressure of 26 randomly selected males in the age group 25 to 30 are shown in the table below. Assume that weight and blood pressure are jointly normally distributed.

Subject	Weight	Systolic BP	Subject	Weight	Systolic BP
1	165	130	14	172	153
2	167	133	15	159	128
3	180	150	16	168	132
4	155	128	17	174	149
5	212	151	18	183	158
6	175	146	19	215	150
7	190	150	20	195	163
8	210	140	21	180	156
9	200	148	22	143	124
10	149	125	23	240	170
11	158	133	24	235	165
12	169	135	25	192	160
13	170	150	26	187	159

- (a) Find a regression line relating systolic blood pressure to weight.
(c) Estimate the correlation coefficient.
(d) Test the hypothesis that $\rho = 0$, using $\alpha = 0.05$.

Answer

(a) $\hat{y} = 69.1044 + 0.4194x$

(c) 0.7735

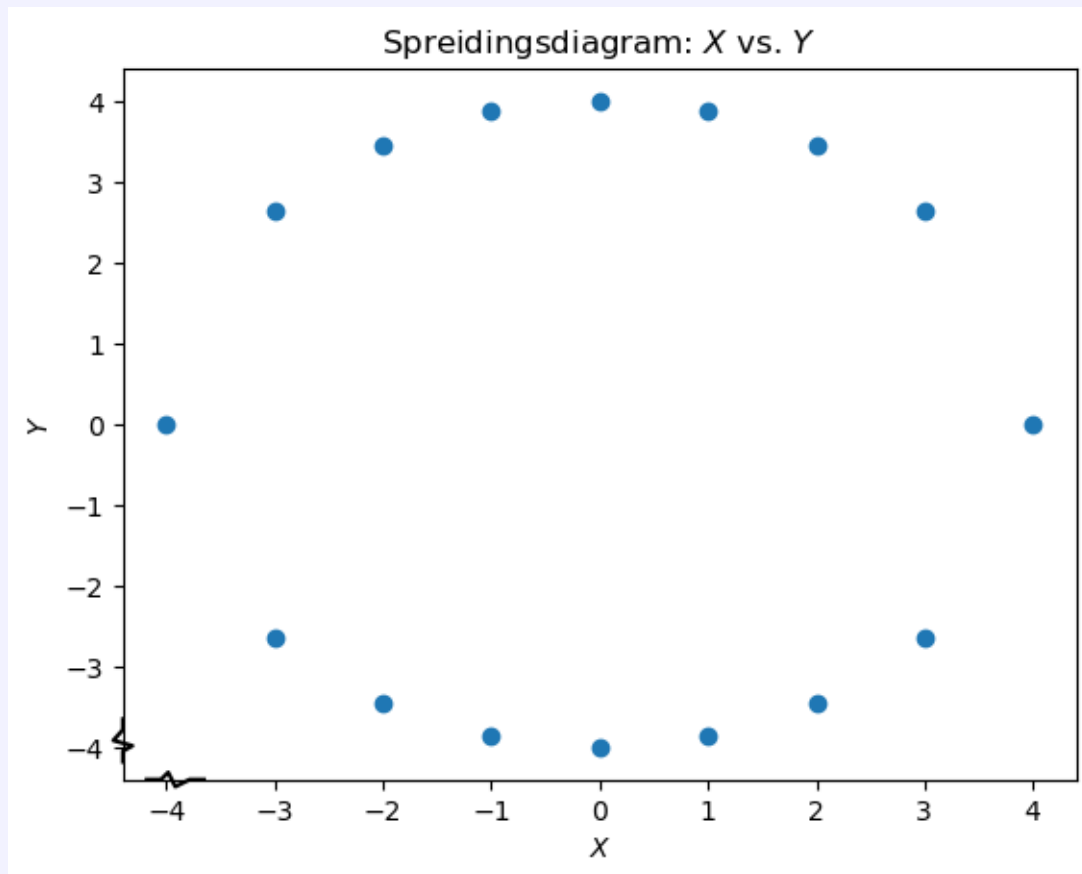
(d) $t_0 = 5.9786 > C_t = 2.0639, p = 0.0000 \leq \alpha = 0.05 \implies \text{reject } H_0$

Exercise 11.76: Consider the (x, y) data in the table below. Calculate the correlation coefficient. Graph the data and comment on the relationship between x and y . Explain why the correlation coefficient does not detect the relationship between x and y .

x	y	x	y
-4	0	0	-4
-3	-2.65	1	3.87
-3	2.65	1	-3.87
-2	-3.46	2	3.46
-2	3.46	2	-3.46
-1	-3.87	3	2.65
-1	3.87	3	-2.65
0	4	4	0

Answer

$r = 0,$



The correlation coefficient does not detect the relationship between x and y because the relationship is not linear.

Exercise 11.88: The table below, adapted from Montgomery, Peck, and Vining (2006), presents the number of certified mental defectives per 10,000 of estimated population in the United Kingdom (y) and the number of radio receiver licenses issued (x) by the BBC (in millions) for the years 1924 through 1937. Fit a regression model relating y and x . Comment on the model. Specifically, does the existence of a strong correlation imply a cause-and-effect relationship?

Year	y	x	Year	y	x
1924	8	1.450	1931	16	4.620
1925	8	1.960	1932	18	5.499
1926	9	2.290	1933	19	6.260
1927	10	2.483	1934	20	7.012
1928	11	2.730	1935	21	7.620
1929	11	3.093	1936	22	8.131
1930	12	3.674	1937	23	8.593

Answer

$\hat{y} = 4.5067 + 2.2152x$, $r = 0.9921$, strong/significant correlation. The existence of a significant correlation does not imply a cause-and-effect relationship.